This report proposes the integration of the RP LiDAR A2M8 into the existing robotic platform in order to extend its perception capabilities. The intention of this integration is to support environment mapping and provide an additional safety-related signal, while keeping the current control structure and behavior unchanged.

The LiDAR is mounted at a height of approximately 14 cm above the ground and performs a two-dimensional scan of the environment at this height. As a result, the sensor captures objects that intersect this horizontal plane, such as walls, furniture legs, or other mid-height obstacles. Objects that are significantly lower than the scan height may not be detected. This limitation is acceptable, as the ultrasonic sensor already covers close-range detection near the ground.

A primary use of the LiDAR is the recording of spatial information during experimental runs. Using the existing ROS and catkin setup, scan data can be logged continuously, allowing the structure of test environments to be documented. These recordings enable later inspection and comparison between experiments conducted in different locations or at different times, which supports consistency and reproducibility.

In addition to mapping, the LiDAR can be used to derive a simple proximity condition for safety support. For example, the system can monitor whether any object is detected within a predefined radius around the robot, such as 30 cm. If this condition is met, a stop signal can be issued to the motion system. This mechanism is intentionally kept simple, as the control system only needs to respond to a binary state indicating the presence or absence of nearby obstacles.

The proposed approach maintains a clear separation between perception and control. The LiDAR operates independently and provides spatial information and optional safety feedback without influencing the existing motion logic directly. The robot can therefore continue to operate using its current sensors even if the LiDAR data is unavailable.

From an implementation perspective, the required software infrastructure is already in place. ROS Melodic and a configured catkin workspace are available, and the LiDAR driver has been successfully tested on the Jetson platform. The remaining work mainly involves configuring the LiDAR node, enabling data recording or mapping, and integrating the simple proximity check. Based on this, the expected implementation effort is estimated to be approximately one to two weeks of part-time work.

This integration provides a practical extension of the platform with minimal risk and complexity. It improves environmental awareness and experimental documentation while preserving the simplicity and reliability of the existing system.